OPERATIONAL SENSE

Database Architecture & Entity Relationship Design

M04 | Logical Design Package | Version 1.2

Document control	Value
Workstream	Operational Sense — EPIC 02
Owner	Ashi Jain (M04)
Intended users	M05 PostgreSQL Schema & Constraints; M06 Asset Hierarchy & Metadata; Groups 1, 3 and 4
Version / date	1.2 / 24 October 2026
Status	Revised for Group 1 alignment review
Source baseline	Industrial Knowledge Base G1 workbook

1. Purpose and design boundary

This document defines the proposed logical data model for Operational Sense. It identifies entities, attributes, primary and foreign keys, relationship cardinalities, integrity rules and mappings from the G1 workbook. The design separates reusable domain reference data from individual troubleshooting scenarios and from real asset records.

The model supports the current synthetic troubleshooting corpus and leaves explicit extension points for the agreed equipment hierarchy, asset criticality assessments, FMEA records, source traceability, review history and future AI & Retrieval metadata.

In scope

- Logical entities and key attributes; primary-key and foreign-key intent.
- Relationships, cardinality and referential-integrity rules.
- Mapping from workbook sheets/fields to proposed database entities.
- Design decisions, assumptions, unresolved dependencies and cross-group review requests.

Outside this deliverable

- PostgreSQL DDL, migrations, indexes, triggers and production loading scripts (M05 implementation).
- Production asset onboarding and final hierarchy seed records (M06 / Group 1).
- Embedding generation, vector search, API, UI and deployment.
- Domain approval of synthetic scenarios, failure taxonomy, criticality scoring or FMEA ratings.

2. Source workbook audit

This design is based on the supplied G1 workbook. The counts below are reproducible checks of the workbook structure; they do not indicate that the technical content has been approved.

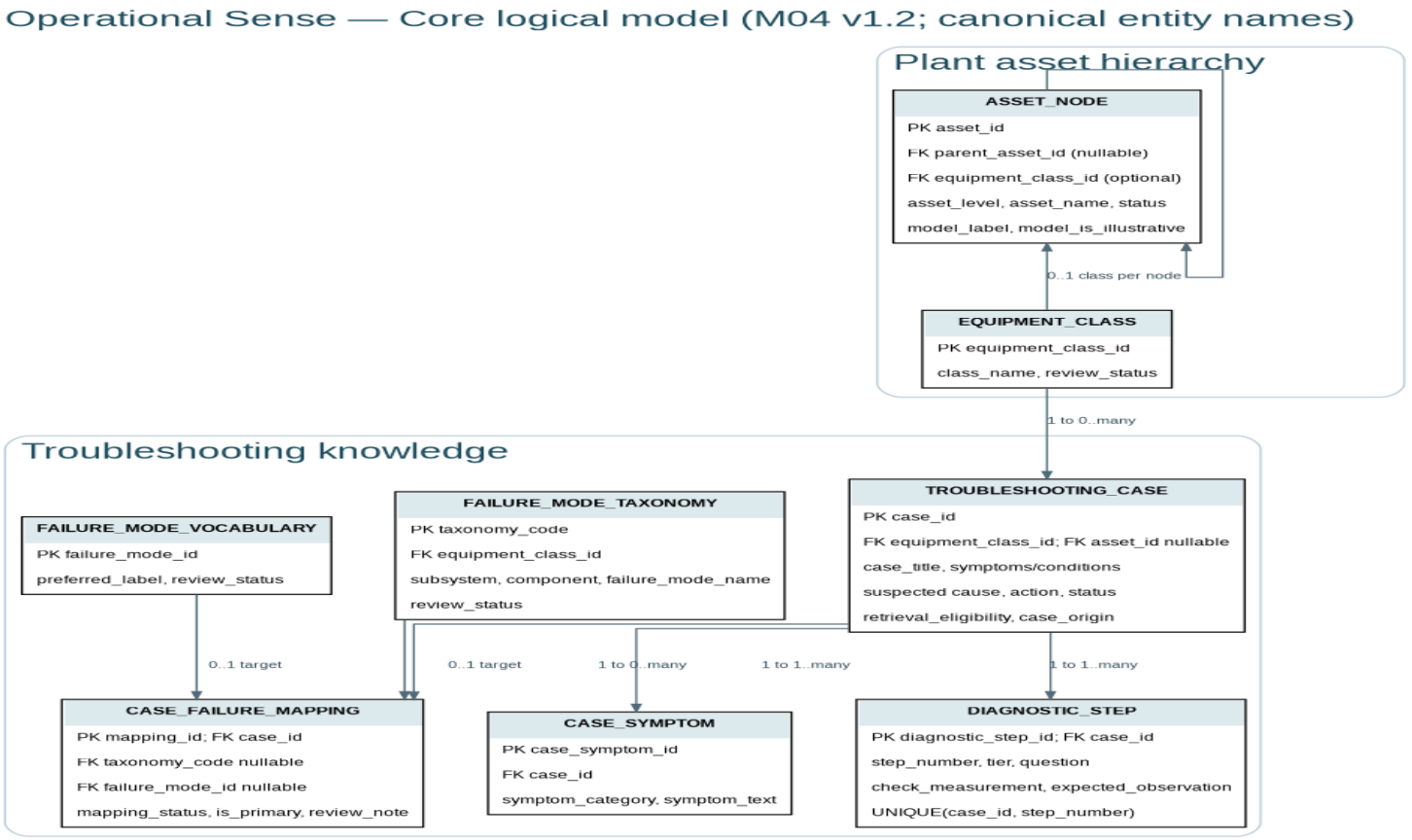
Check	Observed result	Design consequence
Worksheets	18	18 worksheets in the workbook.
Troubleshooting cases	50	50 rows in Import_View; IDs are unique in this snapshot.
Case distribution	17 centrifugal pump; 17 three-phase induction motor; 16 rotary screw air compressor cases	The 50 cases cover three equipment classes.
Diagnostic steps	350	350 rows in Diagnostic_Steps; IDs are unique in this snapshot.
Taxonomy gaps	11	Source quality report flags 11 cases without a taxonomy code.
Ambiguous mappings	13	Mapping_Validation flags 13 ambiguous mappings.
Expert validation	50 pending	Verification_Records has pending overall decisions; no reviewer sign-off is recorded.
Source URL verification	Not run	Data_Quality_Report says URL validation was not performed.

Data handling note: The source workbook identifies all troubleshooting cases as synthetic scenarios. Equipment models are illustrative; do not load these examples as real plant assets or treat suspected causes as confirmed root causes.

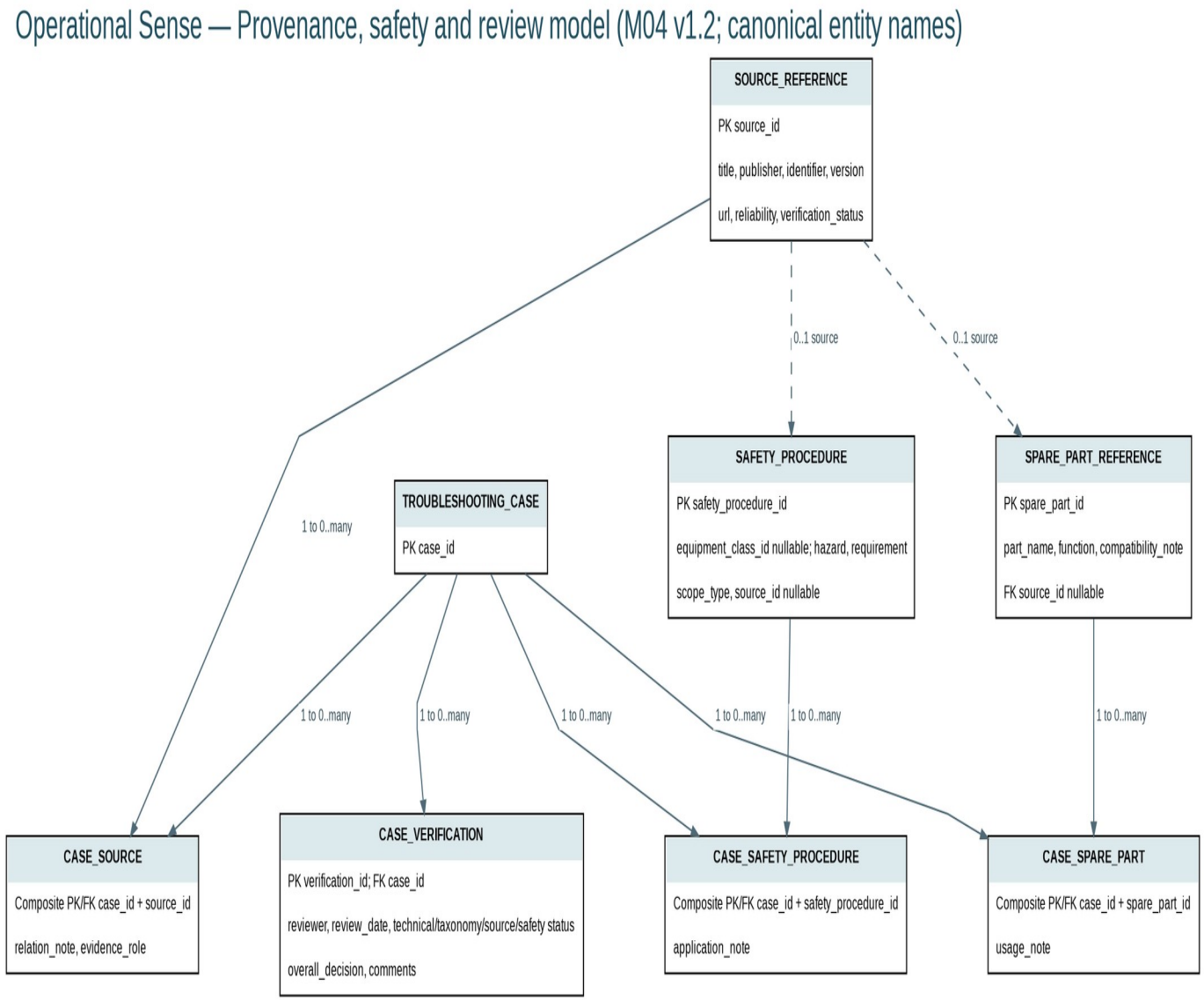
3. Logical model overview

The model is organized into asset hierarchy; troubleshooting cases and diagnostic steps; failure taxonomy and case mappings; source/reference data; and review/assessment records. The ERD views are split for legibility. Relationship cardinalities and join rules are defined in Section 5 and the Relationships worksheet.

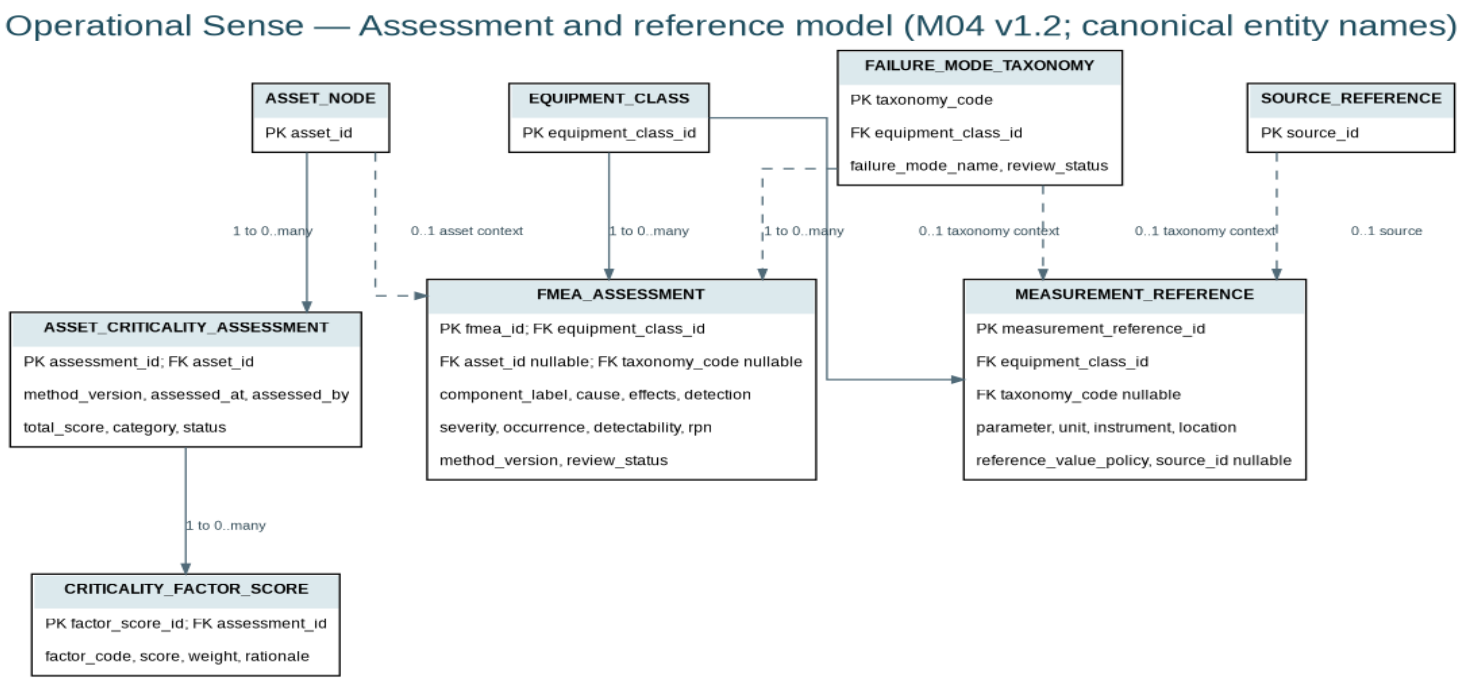
3.1 Core case and hierarchy model



3.2 Source, safety and review relationships



3.3 Asset criticality, FMEA and measurement relationships



4. Entity catalogue

Entity	Responsibility	Primary key	Source / status
EQUIPMENT_CLASS	Controlled equipment-class list	equipment_class_id	Equipment_Reference; Import_View
ASSET_NODE	Plant/Area/System/Equipment/ Component hierarchy node	asset_id	No populated asset register in workbook; future hierarchy
ASSET_CRITICALITY_ASSESSMENT	Versioned criticality assessment for an asset	assessment_id	Asset Criticality Framework dependency; not populated in workbook
CRITICALITY_FACTOR_SCORE	One scored factor within an assessment	factor_score_id	Factor scales/weights pending Group 1 approval
TROUBLESHOOTING_CASE	One troubleshooting scenario, synthetic or later validated	case_id	Import_View; Troubleshooting_Cases
CASE_SYMPTOM	Symptom records associated with a case	case_symptom_id	Import_View symptom fields; controlled vocabulary pending
DIAGNOSTIC_STEP	Ordered diagnostic/check step within a case	diagnostic_step_id	Diagnostic_Steps
FAILURE_MODE_TAXONOMY	Domain taxonomy for equipment/component failure modes	taxonomy_code	Failure_Mode_Taxonomy; Taxonomy_Proposed
FAILURE_MODE_VOCABULARY	Controlled project vocabulary of failure-mode labels	failure_mode_id	Taxonomy_FM_Vocabulary
CASE_FAILURE_MAPPING	Reviewable link from a case to taxonomy/vocabulary concepts	mapping_id	Mapping_Validation; Import_View
FMEA_ASSESSMENT	FMEA assessment for a failure mode in a class/asset/component context	fmea_id	FMEA framework; not a populated case-level table in workbook
SOURCE_REFERENCE	A source, manual, standard or supporting document	source_id	Sources
CASE_SOURCE	Many-to-many link between cases and sources	case_id + source_id	Source IDs in Import_View; Sources
CASE_VERIFICATION	A dated review event for a case	verification_id	Verification_Records; allow history rather than overwrite
MEASUREMENT_REFERENCE	Parameter, unit, instrument and applicability guidance	measurement_reference_id	Measurement_Reference
SPARE_PART_REFERENCE	Reference information for spare parts	spare_part_id	Spare_Parts_Reference
CASE_SPARE_PART	Many-to-many link between cases and spare parts	case_id + spare_part_id	Spare_Parts_Reference; Import_View
SAFETY_PROCEDURE	Reusable safety requirement/procedure	safety_procedure_id	Safety_Procedures
CASE_SAFETY_PROCEDURE	Many-to-many link between cases and safety procedures	case_id + safety_procedure_id	Safety_Procedures; Import_View

5. Relationship and cardinality rules

Cardinality describes how many records may participate in a relationship. “1 : 0..many” means one parent may have zero or many children; each child points back to one parent through its foreign key unless explicitly marked optional.

Relationship	Cardinality	FK / join	Rule
EQUIPMENT_CLASS → ASSET_NODE	1 : 0..many	ASSET_NODE.equipment_class_id	Classifies equipment-level nodes; null for non-equipment hierarchy levels unless approved otherwise.
ASSET_NODE → ASSET_NODE	parent 0..1 : child 0..many	ASSET_NODE.parent_asset_id	Recursive parent-child hierarchy. Enforce Plant→Area→System→Equipment →Component after Group 1 approval.
EQUIPMENT_CLASS → TROUBLESHOOTING_CASE	1 : 0..many	TROUBLESHOOTING_CASE.equipment_class_id	Each case has exactly one equipment class in the current corpus.

Relationship	Cardinality	FK / join	Rule
ASSET_NODE → TROUBLESHOOTING_CASE	1 : 0..many (case link optional)	TROUBLESHOOTING_CASE.asset_id nullable	Link only when a case genuinely concerns an identified real asset; synthetic scenarios remain unlinked by default.
TROUBLESHOOTING_CASE → CASE_SYMPTOM	1 : 1..many	CASE_SYMPTOM.case_id	A case can have multiple symptom entries; current import also retains concise symptom text.
TROUBLESHOOTING_CASE → DIAGNOSTIC_STEP	1 : 1..many	DIAGNOSTIC_STEP.case_id; UNIQUE(case_id, step_number)	Ordered diagnostic workflow; 350 rows currently exist across 50 cases.
TROUBLESHOOTING_CASE → CASE_FAILURE_MAPPING	1 : 0..many	CASE_FAILURE_MAPPING.case_id	Allows mappings to be provisional, ambiguous, or missing without fabricating a taxonomy link.
FAILURE_MODE_TAXONOMY → CASE_FAILURE_MAPPING	0..1 target per mapping	CASE_FAILURE_MAPPING.taxonomy_code nullable	Taxonomy target may be null while a known taxonomy gap is under review.
FAILURE_MODE_VOCABULARY → CASE_FAILURE_MAPPING	0..1 target per mapping	CASE_FAILURE_MAPPING.failure_mode_id nullable	Project-local vocabulary ID is not an ISO code; mapping must remain separately reviewable.
EQUIPMENT_CLASS → FAILURE_MODE_TAXONOMY	1 : 0..many	FAILURE_MODE_TAXONOMY.equipment_class_id	A class can have many applicable taxonomy entries.
TROUBLESHOOTING_CASE → CASE_SOURCE	1 : 0..many	CASE_SOURCE.case_id	Bridge table supports multiple sources per case.
SOURCE_REFERENCE → CASE_SOURCE	1 : 0..many	CASE_SOURCE.source_id	A source may support multiple cases.
TROUBLESHOOTING_CASE → CASE_VERIFICATION	1 : 0..many	CASE_VERIFICATION.case_id	Versioned review history; no review is treated as approval.
ASSET_NODE → ASSET_CRITICALITY_ASSESSMENT	1 : 0..many	ASSET_CRITICALITY_ASSESSMENT.asset_id	An asset may be assessed repeatedly over time or methodology versions.
ASSET_CRITICALITY_ASSESSMENT → CRITICALITY_FACTOR_SCORE	1 : 1..many	CRITICALITY_FACTOR_SCORE.assessment_id	Factor scores are separate so approved factors/weights can evolve without changing the table structure.
EQUIPMENT_CLASS → FMEA_ASSESSMENT	1 : 0..many	FMEA_ASSESSMENT.equipment_class_id	Class-level FMEA can exist without a live asset; asset-specific assessment is optional.
FAILURE_MODE_TAXONOMY → FMEA_ASSESSMENT	0..1 : 0..many	FMEA_ASSESSMENT.taxonomy_code nullable	Use only after taxonomy mapping is approved.
EQUIPMENT_CLASS → MEASUREMENT_REFERENCE	1 : 0..many	MEASUREMENT_REFERENCE.equipment_class_id	Measurement reference is reusable guidance, not a case-specific observed measurement.
SOURCE_REFERENCE → MEASUREMENT_REFERENCE	0..1 source per reference	MEASUREMENT_REFERENCE.source_id nullable	Retain provenance for measurement guidance where a source is supplied.
TROUBLESHOOTING_CASE → CASE_SPARE_PART	1 : 0..many	CASE_SPARE_PART.case_id	Bridge supports multiple spare parts per case.
SPARE_PART_REFERENCE → CASE_SPARE_PART	1 : 0..many	CASE_SPARE_PART.spare_part_id	Spare part may be referenced by multiple cases.
TROUBLESHOOTING_CASE → CASE_SAFETY_PROCEDURE	1 : 0..many	CASE_SAFETY_PROCEDURE.case_id	Bridge links each case to reusable safety procedures.
SAFETY_PROCEDURE → CASE_SAFETY_PROCEDURE	1 : 0..many	CASE_SAFETY_PROCEDURE.safety_procedure_id	A safety procedure may apply to multiple cases.
SOURCE_REFERENCE → SPARE_PART_REFERENCE	0..1 source per part	SPARE_PART_REFERENCE.source_id nullable	Preserve part-reference provenance when available.
SOURCE_REFERENCE → SAFETY_PROCEDURE	0..1 source per procedure	SAFETY_PROCEDURE.source_id nullable	Prefer a traceable source for each reusable safety requirement.

6. Primary keys, foreign keys and integrity rules

- Use stable source case_id values as TROUBLESHOOTING_CASE primary keys for this seed corpus; do not regenerate IDs on each import.
- Use diagnostic_step_id as DIAGNOSTIC_STEP primary key and enforce UNIQUE(case_id, step_number). A diagnostic step must reference an existing case.

- Use controlled EQUIPMENT_CLASS records; enforce uniqueness on equipment_class_id and, after domain review, on canonical class_name.
- Use ASSET_NODE.parent_asset_id as a self-referencing foreign key. Validate allowed parent-child levels in a constraint or trigger only after Group 1 confirms the hierarchy contract.
- Keep taxonomy_code and failure_mode_id as distinct foreign keys. The project vocabulary ID is not automatically an ISO 14224 code.
- Allow a nullable taxonomy target only when mapping_status and a documented gap/review reason explain the missing link; do not silently discard the gap.
- Use bridge tables for case-to-source, case-to-spare-part and case-to-safety-procedure many-to-many relationships. Composite primary keys should prevent duplicate links.
- Model review as dated CASE_VERIFICATION events so later reviews do not overwrite prior decisions. A blank reviewer/date and PENDING status is not approval.
- Keep criticality and FMEA scoring method versions with each assessment. Store the calculated RPN only after the rating scales and calculation rule are approved.
- Apply ON DELETE RESTRICT for referenced domain/master records by default. Case-child records may use ON DELETE CASCADE only where the team explicitly agrees that child records have no independent retention requirement.
- No numeric alarm threshold should be inferred from generic text; thresholds require an approved OEM specification or site baseline and applicability conditions.

7. Source-to-entity mapping

The accompanying editable workbook contains the complete field-by-field mapping for all Import_View columns and relevant fields from Diagnostic_Steps, Sources, Verification_Records, Measurement_Reference, Spare_Parts_Reference and Safety_Procedures. The summary below shows the primary mappings.

Workbook field(s)	Destination	Mapping rule
Import_View.case_id	TROUBLESHOOTING_CASE.case_id	Direct; preserve source ID.
equipment_class / equipment_class_id	EQUIPMENT_CLASS	Controlled lookup; label and ID must agree.
symptom_category / symptom_reported_hinglish / symptom_structured	CASE_SYMPTOM	Retain original symptom text and structured description.
Tier 1 / Tier 2 / Tier 3 narrative fields	TROUBLESHOOTING_CASE and DIAGNOSTIC_STEP	Keep case summary and ordered step detail distinct.
taxonomy_code / fm_id_vocab / mapping_status	CASE_FAILURE_MAPPING + taxonomy/vocabulary lookups	Keep mapping status, gap and review notes.
source_ref_ids + Sources sheet	CASE_SOURCE + SOURCE_REFERENCE	Preserve reference identity and verification status.
Diagnostic_Steps	DIAGNOSTIC_STEP	Transform source labels to consistent snake_case fields and enforce step ordering.
Verification_Records	CASE_VERIFICATION	Load pending states as pending; never fabricate reviewer approval.
Measurement_Reference	MEASUREMENT_REFERENCE	Keep units, applicability and reference-value policy.
Spare_Parts_Reference / Safety_Procedures	Reference entities + case bridge tables	Normalize reusable reference records and preserve associations.

8. Field and type guidance

The workbook mapping records logical destinations. Physical PostgreSQL types and nullability should be finalized with M05 against the approved Common Data Model. The following are recommended starting types, not a substitute for that review.

Field group	Suggested PostgreSQL type	Constraint / rationale
Identifier fields	TEXT or VARCHAR with explicit length only if agreed	Preserve external IDs such as CASE-PUMP-001 exactly.
Human-readable names and labels	TEXT	Use controlled lookup rows where values must be standardized.
Long explanations / diagnostic instructions	TEXT	Avoid truncation; do not store large narratives in short VARCHAR columns.
Review / lifecycle status	TEXT + CHECK or lookup table	Allowed values must be agreed and documented.
Boolean flags	BOOLEAN	Use actual boolean values, not “Yes/No” strings.
Review dates / event timestamps	DATE / TIMESTAMPTZ	Agree whether date-only or timestamp with timezone is required.
Criticality scores / FMEA ratings	SMALLINT / NUMERIC	Bounds and weights depend on approved rating framework.
RPN	INTEGER or generated calculation	Use only after the S × O × D rule and rating scale are approved.
Flexible source metadata (if unavoidable)	JSONB sparingly	Do not use JSONB as a shortcut for core relationships or filterable fields.

9. Data-quality and ingestion controls

- Pre-load: verify workbook version, required headers, 50 case IDs, unique IDs, allowed equipment classes and source-ID existence.
- Pre-load: validate every diagnostic step references a known case; validate step numbers are positive and unique per case.
- Pre-load: keep original/source wording in provenance fields where normalization changes labels.
- Load: use a transaction and report rejected rows; do not silently coerce malformed values or fill unknown technical facts.
- Post-load: compare row counts to source counts, check foreign-key orphans, verify case-to-step counts and confirm taxonomy gaps remain visible.
- Operational: distinguish structural validity from technical validation. A record can pass field checks and still require expert review.
- Retrieval: keep retrieval_eligibility explicit; synthetic and unreviewed cases should be excluded from any production-facing recommendation path unless the project explicitly approves a labelled demo mode.

9A. Group 1 hierarchy and taxonomy alignment

The Group 1 hierarchy proposal (Draft v0.1) is treated as a review input, not an approved contract. It distinguishes equipment class, equipment type, subsystem, component and failure mode. This is a knowledge taxonomy and must not be conflated with the plant asset hierarchy (Plant → Area → System → Equipment → Component). The two structures should be linked through reviewed mappings rather than forced into a single parent-child chain.

Design treatment for this revision

- Keep ASSET_NODE for the plant asset hierarchy and its site-specific parent-child relationships.
- Keep EQUIPMENT_CLASS / equipment-type references and failure taxonomy as controlled knowledge reference data; map cases and components to them using explicit foreign keys or bridge tables.
- Keep project taxonomy_code separate from failure_mode_id. Do not label project codes as ISO 14224 codes unless a reviewed crosswalk explicitly supports that claim.

- Preserve source identifiers and labels. Do not silently rename taxonomy codes, merge subsystem labels, split composite components or invent missing IDs during ingestion.
- Treat the 11 entries marked proposed in the Group 1 proposal as unapproved; retain their proposal status and owner/review notes.
- Keep synthetic case records and unreviewed mappings ineligible for production recommendations unless a clearly labelled demo mode is explicitly approved.

Open alignment items

Confirm the equipment class/type boundary, proposed subsystem merges, motor taxonomy-code convention, composite component names, pump-versus-driver boundary, and ambiguous case mappings with M01/M02 and Groups 2–3. Confirm structured retrieval filters and metadata with Group 3, and physical schema constraints with M05. No approval is implied by this revision.

9B. Model consistency and revision control

The Entity_Inventory and Relationships worksheets in M04_Entity_Relationship_Reference_v1.2.xlsx are the canonical logical-model reference for this package. The three ERD views and editable sources have been reconciled to use the same entity names and key relationships. Where a decision remains open, the diagram shows the proposed logical relationship and does not imply domain approval.

Canonical naming corrections in this revision include CASE_FAILURE_MAPPING (with separate optional taxonomy_code and failure_mode_id references), FMEA_ASSESSMENT, and MEASUREMENT_REFERENCE. FMEA_ASSESSMENT supports equipment-class context with optional asset and taxonomy context; MEASUREMENT_REFERENCE is equipment-class scoped, with optional taxonomy and source references. These are reference/assessment records, not assertions that real plant measurements or approved FMEA ratings have been supplied.

Version control: the report, reference workbook, ERD exports and editable sources are packaged as M04 v1.2. The exact Common Data Model and JSON Schema versions remain an open cross-group dependency until the approved artifacts and sign-off are recorded. No Common Data Model version is represented as approved by this package.

10. Assumptions and decisions requiring sign-off

ID	Owner	Decision required	Why it matters	Status
Q01	Group 1 + M06	Confirm canonical hierarchy levels and whether “Subsystem” is a formal node or descriptive metadata.	Blocks parent-level validation and ASSET_NODE constraints.	Open
Q02	Groups 1 + 3 + 4	Provide the agreed Common Data Model artifact and record its exact version plus JSON Schema version; approval must be confirmed by Groups 1, 3 and 4.	Blocks final field contract and acceptance sign-off. M04 does not infer or claim an approved version.	Open
Q03	Group 3	Confirm structured retrieval filters, embedding text fields, metadata contract and eligibility rules.	Prevents mismatch with AI & Retrieval pipeline.	Open
Q04	M05	Confirm PostgreSQL type conventions, naming, delete/update behavior, indexes and migration constraints.	Needed before physical schema implementation.	Open

ID	Owner	Decision required	Why it matters	Status
Q05	M06 + Group 1	Confirm whether current plant/facility/production-line fields are fictional context or should map to asset hierarchy nodes.	Avoids turning synthetic case context into real assets.	Open
Q06	Group 1 / domain steward	Approve failure taxonomy and project-local vocabulary mappings; resolve ambiguous mappings and taxonomy gaps.	11 taxonomy gaps and 13 ambiguous mappings are reported by source workbook.	Open
Q07	Group 1 + M05	Confirm criticality scoring factors, scales, weightages, and whether assessment history is required in v1.	Assessment schema is intentionally method-versioned.	Open
Q08	Group 1 + domain reviewer	Confirm FMEA RPN method, scales and interpretation rules before storing calculated RPN.	Avoids embedding unapproved risk scoring logic.	Open
Q09	M05 + M06	Confirm identifier strategy for asset nodes and hierarchy uniqueness rules (global vs parent-scoped).	Affects PKs and UNIQUE constraints.	Open
Q10	Groups 1 + 3 + 4	Record cross-group decisions and formal approval before marking architecture Approved.	Required by Epic 2 dependency and acceptance criteria.	Open

11. Cross-group handoff

Reviewer	Required feedback
Group 1 — Domain Research	Confirm hierarchy levels, naming conventions, taxonomy labels, failure-mode mappings and criticality/FMEA frameworks.
Group 3 — AI & Retrieval	Confirm metadata filters, embedding text fields, retrieval eligibility contract and schema requirements.
Group 4 — QA / Integration / DevOps	Confirm consistency checks, test evidence, versioning and integration acceptance evidence.
M05 — PostgreSQL Schema & Constraints	Review entity boundaries, PostgreSQL types, PK/FK rules, unique constraints, delete behavior and migration feasibility.
M06 — Asset Hierarchy & Metadata	Confirm asset identifiers, hierarchy-node semantics, component representation and uniqueness scope.

Approval record: No formal sign-off has been supplied. Record each reviewer, decision, date and agreed change in the shared issue or pull request before marking this design approved.

Implementation approval gate: Do not mark M04 Approved until the Common Data Model version, hierarchy contract, taxonomy mappings, retrieval metadata and scoring rules have named owners and recorded decisions. M04 v1.2 is a reconciled review package, not a substitute for those approvals.

12. M04 completion checklist

Deliverable / checkpoint	Status
Entity inventory and purposes	Included
PK/FK intent and relationship cardinality	Included
ERD images and editable source	Included
Workbook-to-entity field mapping	Included in companion workbook
Source-data audit and known quality gaps	Included
Assumptions and cross-group questions	Included
PostgreSQL implementation	M05 responsibility; not part of M04 design artifact

Deliverable / checkpoint	Status
Formal Common Data Model approval	Pending relevant group review

Appendix A. Terms used in this document

Term	Definition
Entity	A category of information stored as a database table.
Attribute	A field/column describing an entity.
Primary key (PK)	A field or field combination that uniquely identifies a row.
Foreign key (FK)	A reference to a row in another table, used to maintain valid connections.
Cardinality	The allowed number of records participating in a relationship.
Bridge table	A table used to connect two entities when each can relate to many records of the other.
Synthetic scenario	An illustrative case, not evidence of an actual industrial incident.
RPN	Risk Priority Number; typically derived from Severity × Occurrence × Detection, subject to the approved FMEA method.
LOTO	Lockout/tagout: an energy-isolation procedure. Site-approved safety procedures take precedence.
Retrieval eligibility	A rule indicating whether a case is allowed into the retrieval/recommendation flow.